PX series

Close-coupled hygienic centrifugal pumps, 5 to 420 m^3/h



10 sizes
one hydraulic family

420 m^3/h
maximum flow

10 bar
maximum differential pressure

140 °C
maximum product temperature

Overview

The PX is a close-coupled centrifugal pump for hygienic and light process duties. The hydraulic parts are machined in one setup, so the impeller runs with small clearances and little recirculation: a lower power draw than open-impeller pumps of the same size, and a flat efficiency curve around the best point.

A compression coupling fixes the stub shaft to the motor shaft and keeps the alignment without shims. The semi-open impeller with swept vanes moves the product with little shear.

Where it is used

Dairy, brewing, beverage, food and personal care plants where the product must not be sheared and the pump must be back in service quickly after cleaning. The PX is made for cleaning-in-place (CIP) and sterilisation-in-place (SIP): drainable casing, no dead legs, verified CIP cleanability.

Ten sizes cover capacities up to 420 m^3/h and differential pressures up to 10 bar at 50 Hz, so a duty is met by one pump near its best point rather than two in series or parallel.

Design

All product-wetted steel parts (casing, impeller, impeller nut and back plate) are EN 1.4404 (AISI 316L). A stainless shroud covers the motor and four adjustable stainless legs carry the unit.

The standard seal is a single mechanical shaft seal; a flushed single seal and a double mechanical seal are options. The seal loads from the front, with the spring and washers on the atmospheric side, so a seal change takes minutes without removing the pipework, and the balanced design holds through pressure shocks.

01 The range at a glance

Pump model	PX-5	PX-10	PX-15	PX-20	PX-30	PX-42	PX-55	PX-70	PX-85	PX-100
Max flow (m^3/h)	12	25	40	60	90	130	180	240	320	420
Max head (m)	24	40	36	58	46	72	62	100	84	48
Motor range (kW)	0.75–2.2	1.5–4.0	1.5–7.5	3.0–18.5	3.0–22	4.0–45	5.5–75	11–75	22–110	30–110
Inlet	DN40	DN65	DN65	DN80	DN80	DN100	DN100	DN125	DN150	DN150
Outlet	DN32	DN50	DN50	DN65	DN65	DN80	DN80	DN100	DN125	DN125

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02 Technical data

Materials

Product wetted steel parts	EN 1.4404 (AISI 316L)
Other steel parts	Stainless steel
Inside surface finish	Ra $\leq$ 1.6 μm, glass-bead blasted
Product wetted elastomers	EPDM
Rotary seal face	Carbon
Stationary seal face	Silicon carbide

Motor sizes

50 Hz	0.75 – 110 kW
60 Hz	0.75 – 110 kW

Min/max motor speed

2 poles, 0.75 – 45 kW	900 – 4000 rpm
2 poles, 55 – 110 kW	900 – 3600 rpm
4 poles, 0.75 – 75 kW	900 – 2200 rpm

Motor

Foot-flanged IEC motor. 2 poles = 3000/3600 rpm at 50/60 Hz, 4 poles = 1500/1800 rpm at 50/60 Hz. IP 55 with drain hole and labyrinth plug, insulation class F, efficiency class IE3.

Warranty

Three years on PX pumps, covering all non-wear parts, on the condition that Norrhavn service parts are used.

Noise

Sound pressure at 1 m, 2 poles, 50 Hz	Max 78 dB(A) to PX-55, max 84 dB(A) from PX-70

03 Operating data

Max inlet pressure

PX-5	600 kPa (6 bar)
PX-10 – PX-70	1000 kPa (10 bar)
PX-70, 60 Hz	500 kPa (5 bar)
PX-85 – PX-100	500 kPa (5 bar)

Temperature

Temperature range	-10 °C to +140 °C (EPDM)
Flush media	Max 70 °C
Flush housing sterilisation (pump not in operation)	Max 125 °C

Flushed shaft seal

Water pressure inlet	Max 1 bar
Water consumption	0.25 – 0.5 l/min

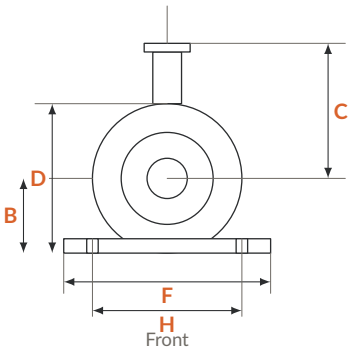
Double mechanical shaft seal

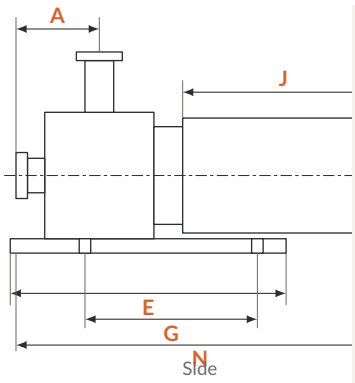
Water pressure inlet, PX-5 to PX-55	Max 500 kPa (5 bar)
Water pressure inlet, PX-70 to PX-100	Max 300 kPa (3 bar)
Water consumption	0.25 – 0.5 l/min

Seal flush connections

PX-5 – PX-70, PX-100	1/8" G
PX-85	Ø6 tube

04 Dimensions





Letter key

A	Suction flange face to discharge axis
B	Shaft centre height above base
C	Discharge flange face above base
D	Casing outer diameter
E	Baseplate length
F	Baseplate width
G	Foot bolt spacing, length
H	Foot bolt spacing, width
J	Motor length, flange to fan cover
K	Motor height over shaft centre
N	Overall length, suction flange face to fan cover

Pump specific measures (mm)

Pump model	PX-5	PX-10	PX-15	PX-20	PX-30	PX-42	PX-55	PX-70	PX-85	PX-100
A	80	100	115	130	150	170	185	205	220	230
B	75	95	110	120	140	160	175	190	205	220
C	175	225	260	285	330	375	415	450	485	510
D	120	155	180	200	230	260	285	310	335	355
E	140	180	205	230	265	300	330	360	385	410
F	85	105	125	140	160	180	200	215	230	245
G	110	145	165	185	215	240	265	290	310	330
H	65	85	95	105	125	140	155	170	180	190
Mass (kg)[2]	29	41	48	62	84	112	158	215	290	380

[2] Pump with the smallest standard motor, dry.

Motor specific measures (mm)

Motor frame	IEC80	IEC90	IEC100	IEC112	IEC132	IEC160	IEC180	IEC200	IEC250	IEC280
Motor kW	0.75/1.1	1.5/2.2	3.0	4.0	5.5/7.5	11/15/18.5	22	30/37/45	55/75	90/110
J	262	262	282	285	304	332	352	372	446	496
K	125	157	185	198	196	262	286	399	394	584
N (PX-5 – -15)	555	700	730	765	785	–	–	–	–	–
N (PX-20 – -55)	–	–	840	930	1005	1095	1140	1265	1325	–
N (PX-70 – -100)	–	–	–	–	–	1150	1250	1430	1490	1690

N is given for the largest pump of each group; smaller pumps in the group are shorter by the difference in A.

Motor overview

Pump model	PX-5	PX-10	PX-15	PX-20	PX-30	PX-42	PX-55	PX-70	PX-85	PX-100
Motor range (IEC)	IEC80–IEC90	IEC90–IEC112	IEC90–IEC132	IEC100–IEC160	IEC100–IEC180	IEC112–IEC200	IEC132–IEC250	IEC160–IEC250	IEC180–IEC280	IEC200–IEC280

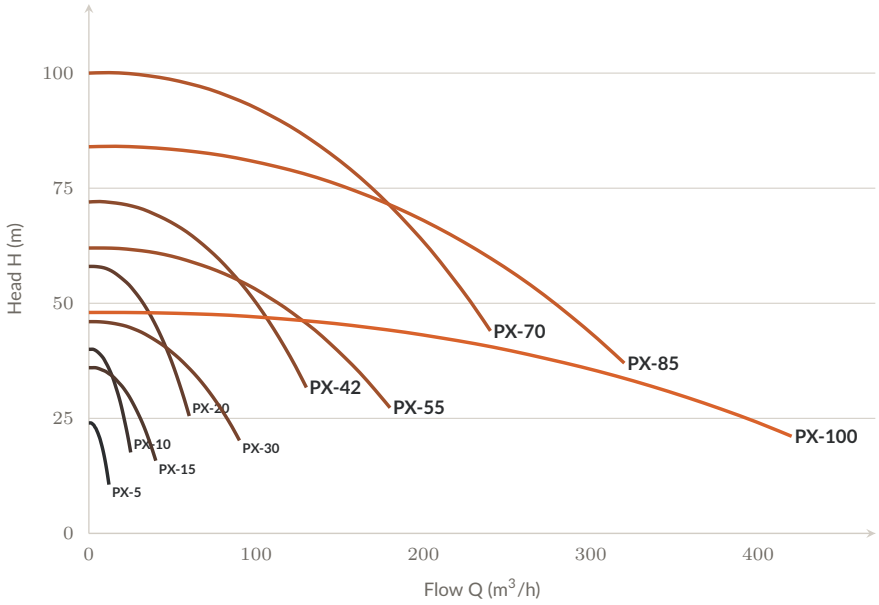
PX-5 to PX-85: 2-pole motors. PX-100: 4-pole motors.

05 Connections

Pump model		PX-5	PX-10 PX-15	PX-20 PX-30	PX-42 PX-55	PX-70	PX-85 PX-100
Clamp ISO 2852	M1	21	21	21	21	21	–
	M2	21	21	21	21	21	–
Union SMS 1145	M1	20	24	24	35	35	–
	M2	20	20	24	24	35	–
Union DIN 11851	M1	22	25	30	30	30	50
	M2	22	22	25	30	30	50
Flange EN 1092-1 PN16	M1	–	–	18	18	20	22
	M2	–	–	18	18	20	22
Inlet[1]	N1	DN40	DN65	DN80	DN100	DN125	DN150
Outlet[1]	N2	DN32	DN50	DN65	DN80	DN100	DN125

[1] Other sizes on request. M1 and M2 are the face-to-centre distances of the inlet and outlet fittings in mm.

06 Performance



Test conditions
Frequency 50 Hz.
Synchronous speed 3000 rpm.
Water at 20 °C.
Full impeller diameter.

Each curve ends at the largest flow for continuous duty. Reduced impeller diameters move a curve down; ask for the selection program for intermediate points.

07 Options

- Impeller with reduced diameter.
- Flushed shaft seal.
- Double mechanical shaft seal.
- Rotating seal face of silicon carbide.
- Product wetted elastomers FKM or FEP.
- 1/2" vertical drain connection.
- Product wetted surface finish polished to Ra $\leq$ 0.8 μm, with or without certificate.
- Inducer (PX-10 to PX-55).
- Adjustable pads under the legs.
- Motor for other voltage and/or frequency.
- Half-speed motor.
- Increased-safety or flameproof motor.
- ATEX execution (PXex).

08 Specifying a pump

State the following with an enquiry or order:

1. Pump size.
2. Connections.
3. Impeller diameter.
4. Motor size.
5. Voltage and frequency.
6. Flow, pressure and temperature.
7. Density and viscosity of the product.
8. Options.

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